TROPIC Simulation Model Analysis Plan

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Current sealed controlled release: v0.3.0-clinical-simulation docs/RELEASE_NOTE_v0.3.0-clinical-simulation.md

Informative annex boundary. This plan is a data-free simulation-science methods evaluation layered on the historical sealed clinical-simulation release. It is not MIDD evidence, a filing artifact, confirmatory efficacy evidence, sponsor approval, or evidence of regulator acceptance.

Document Control

Item	Governed value
Protocol identifier	TROPIC-SIM-MAP-001
Protocol version	1.0.0
Protocol status	FROZEN_MAP
Protocol frozen on	2026-08-14
Full run started after freeze	Yes
Prospectively recorded deviations	None
Governed protocol SHA-256	09798cd52adedd742a10f39266c67df0d9fe40b4c693912e4962b47601446b61
Authoritative result SHA-256	16bfab6f9abc171fa52ddd9d6d7c5035a4502c37956851cc247c51b8bc3d4fb8
Scientific output SHA-256	46ca63556e65c649b17c7ace2b70447691fd50d621c84d630880bb988fdb643e

The result hash above was added by this post-run report build for traceability only. It is not presented as a prospective MAP element; design assumptions, methods, criteria, scenario identities, and seeds are governed by the frozen protocol.

ICH M15 Context and Model Risk

Assessment element	Prespecified statement
Question of interest	Under explicit public-aggregate or stress-test assumptions, what are the rejection probabilities and numerical precision of a one-sided unstratified log-rank test in a simplified TROPIC-like fixed two-arm OS or PFS design?
Context of use	Low-influence internal methods evaluation and reviewer training for simulation engineering, numerical precision, and sensitivity to stated design assumptions.
Model influence	LOW

Assessment element	Prespecified statement
Consequence of a wrong decision	Misinterpreting numerical operating characteristics as clinical evidence could misstate design performance or treatment benefit; the hard use boundary prevents those decisions from relying on this model.
Model risk	MODERATE
Model impact	LOW

Context and Risk Interpretation

- Scientific assumptions are deliberately simplified and not identifiable from authoritative joint IPD. Consequence severity is potentially high, but model influence is constrained to low by governance and explicit non-use controls.
- No authoritative comparator IPD or joint endpoint/covariate distribution.
- Hazards, dropout, discontinuation, delayed effect, and waning are assumptions.
- No independent organizational statistical or medical review.

ICH E9(R1) Estimand

Attribute	Prespecified definition
Population	Hypothetical randomized participants meeting a simplified TROPIC-like analysis population; fixed public-randomized-design calibration of 377 control and 378 experimental participants. This is distinct from the repository's available real MP cohort of N=371 and does not imply that participant-level data are available.
Treatment conditions	control: Synthetic control strategy from randomization through follow-up.; experimental: Synthetic experimental strategy from randomization through follow-up.
Variables / endpoints	OS: Time from randomization to death from any cause.; PFS: Time from randomization to progression or death from any cause.
Population-level summary	Between-arm survival-distribution contrast evaluated by the one-sided intent-to-treat log-rank statistic; repeated-trial summary is rejection probability.

Intercurrent Events

Intercurrent event	Strategy	Simulation / analysis handling
permanent_treatment_discontinuation	TREATMENT_POLICY	Follow-up and endpoint ascertainment continue. Reference scenarios retain the assigned-arm hazard; stress scenarios explicitly allow effect waning after discontinuation without censoring the endpoint.
subsequent_anticancer_therapy	TREATMENT_POLICY	Endpoint follow-up would continue regardless of subsequent therapy; a separate therapy process is not generated in this bounded implementation.
death	OS: OUTCOME_EVENT_NOT_INTERCURRENT; PFS: COMPOSITE	Death is the OS event and is included in the PFS composite event.

Missing Data and Censoring

Mechanism	Classification	Handling
Independent Withdrawal	MISSING_ENDPOINT_OBSERVATION_NOT_INTERCURRENT_EVENT	Independently generated withdrawal censors at last observation. High-dropout scenarios stress this assumption; informative withdrawal is not modeled.
Administrative Censoring	Censoring	Censor at the fixed common analysis date after staggered enrollment.

Missing observations and administrative censoring are handled according to the governed methods; they are not silently relabelled as intercurrent events.

ADEMP and OCTAVE Framework

Objectives and Aims

- Verify a reproducible simulation implementation and characterize operating characteristics; do not reproduce the original TROPIC trial or estimate a clinical treatment effect.
- Estimate Type I error or power and its Monte Carlo uncertainty for the governed scenario grid.
- Verify implementation and characterize fixed-design operating characteristics.

Characteristics and Data-Generating Mechanisms

- Individual enrollment, piecewise-exponential endpoint time, independent withdrawal, optional treatment discontinuation with post-discontinuation waning, and fixed administrative censoring.
- OS/PFS baseline hazards, treatment-effect shape, withdrawal, and discontinuation assumptions.

Trial Design and Analysis Methods

Design element	Governed value
Allocation Basis	Public randomized design calibration (377 control, 378 experimental), distinct from the repository's available real MP N=371; no authoritative participant-level comparator data are used.
Allocation	control: 377; experimental: 378
Alpha One Sided	0.025
Enrollment Months	12
Analysis Month	24
Batch Size	1000
Analysis Method	ONE_SIDED_UNSTRATIFIED_LOGRANK
Omitted Design Feature	Original stratification is not simulated because authoritative joint stratification-factor distributions are unavailable.
Endpoint Control Medians Months	OS: 12.7; PFS: 1.4

- Fixed parallel two-arm allocation with staggered enrollment and common analysis date.
- One-sided unstratified log-rank test at alpha 0.025.
- Intent-to-treat one-sided log-rank analysis.

Valuation Metrics and Evidence

- Rejection numerator, denominator, probability, MCSE, and 95% Wilson interval.
- Completed/failed accounting and failure reasons.
- Mean event and censor proportions by randomized arm.
- Type I error or power, Monte Carlo precision, event/censoring rates, and execution failures.
- Protocol/scenario/code hashes, deterministic trials, analytic benchmark, and machine-readable results.

Parameter Provenance

Provenance class	Governed assumptions
Public Aggregate	TROPIC-like arm sizes and aggregate OS/PFS medians are used only as design-calibration assumptions.; Published-effect hazard ratios are public-aggregate scenario inputs, not re-estimated effects.
Assumed Stress	Withdrawal, delayed-effect, and discontinuation/waning inputs are engineering stress assumptions.
Unavailable	Authoritative joint comparator IPD, endpoint correlations, and intercurrent-event distributions.

Scenario Registry

Scenario	Class	Endpoint	Assumption basis	Varying factors	Replicates	Seed
OS_NULL_REFERENCE	KEY_NULL	OS	ASSUMED_REFERENCE	treatment hr segments=start month: 0; hazard ratio: 1; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0; post discontinuation hazard ratio=1	100000	2026081401

Scenario	Class	Endpoint	Assumption basis	Varying factors	Replicates	Seed
PFS_NULL_HIGH_DROPOUT	KEY_NULL_OPERATIONAL_STRESS	PFS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 1; withdrawal rate 12m=control: 0.25; experimental: 0.25; discontinuation rate 12m=0; post discontinuation hazard ratio=1	100000	2026081402
OS_PUBLISHED_EFFECT	ALTERNATIVE	OS	PUBLIC_AGGREGATE	treatment hr segments=start month: 0; hazard ratio: 0.7; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0; post discontinuation hazard ratio=1	25000	2026081403
PFS_PUBLISHED_EFFECT	ALTERNATIVE	PFS	PUBLIC_AGGREGATE	treatment hr segments=start month: 0; hazard ratio: 0.74; withdrawal rate 12m=control: 0.08; experimental: 0.08; discontinuation rate 12m=0; post discontinuation hazard ratio=1	25000	2026081404

Scenario	Class	Endpoint	Assumption basis	Varying factors	Replicates	Seed
OS_MEDIAN_CALIBRATED	ALTERNATIVE	OS	PUBLIC_AGGREGATE_CALIBRATION	treatment hr segments=start month: 0; hazard ratio: 0.84106; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0; post discontinuation hazard ratio=1	25000	2026081405
PFS_MEDIAN_CALIBRATED	ALTERNATIVE	PFS	PUBLIC_AGGREGATE_CALIBRATION	treatment hr segments=start month: 0; hazard ratio: 0.5; withdrawal rate 12m=control: 0.08; experimental: 0.08; discontinuation rate 12m=0; post discontinuation hazard ratio=1	25000	2026081406
OS_DELAYED_NON_PH	ALTERNATIVE_STRESS	OS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 1; start month: 4; hazard ratio: 0.65; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0; post discontinuation hazard ratio=1	25000	2026081407

Scenario	Class	Endpoint	Assumption basis	Varying factors	Replicates	Seed
PFS_DELAYED_NON_PH	ALTERNATIVE_STRESS	PFS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 1; start month: 1; hazard ratio: 0.6; withdrawal rate 12m=control: 0.08; experimental: 0.08; discontinuation rate 12m=0; post discontinuation hazard ratio=1	25000	2026081408
OS_WANING_AFTER_DISCONTINUATION	ALTERNATIVE_STRESS	OS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 0.7; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0.35; post discontinuation hazard ratio=1	25000	2026081409
PFS_WANING_AFTER_DISCONTINUATION	ALTERNATIVE_STRESS	PFS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 0.74; withdrawal rate 12m=control: 0.08; experimental: 0.08; discontinuation rate 12m=0.45; post discontinuation hazard ratio=1	25000	2026081410

Scenario identifiers, requested replication counts, and seeds are exact governed values. Null, alternative, non-proportional-effect, and operational-stress roles must remain explicit; scenarios are not pooled into an undocumented average.

Monte Carlo and Operating-Characteristic Methods

- One-sided alpha: 0.025
- Batch size: 1000
- One-sided unstratified log-rank test at alpha 0.025.
- Rejection numerator, denominator, probability, MCSE, and 95% Wilson interval.; Completed/failed accounting and failure reasons.; Mean event and censor proportions by randomized arm.

For every binomial operating characteristic, the report presents the exact numerator and denominator, estimate, Monte Carlo standard error, and Wilson 95% interval supplied by the authoritative scientific JSON.

Prespecified Acceptance Criteria

Control	Criterion	Scope
Key Null Min Completed	100000	All applicable scenarios
Alternative Min Completed	25000	All applicable scenarios
Alternative Precision Justification	At 25,000 completed replicates the worst-case binomial MCSE is no greater than sqrt(0.25/25000)=0.003163. Alternative and stress-scenario probabilities are descriptive because no sponsor-approved performance threshold or MCID is available.	All applicable scenarios
Target Mcse Key Null	0.0005	All applicable scenarios
Max Failure Rate	0.001	All applicable scenarios
Analytic Null Probability	0.025	All applicable scenarios
Analytic Null Abs Tolerance Floor	0.001	All applicable scenarios
Analytic Null Mcse Multiplier	3	All applicable scenarios
Wilson Confidence Level	0.95	All applicable scenarios
Null Acceptance	Wilson interval contains alpha and absolute deviation is no greater than max(0.001, 3MCSE); upper Wilson bound is not required to be below alpha.	All applicable scenarios

Execution, Monte Carlo precision, design operating characteristics, and evidence qualification are assessed separately. A successfully executed, precisely estimated unacceptable design remains a design failure or review finding.

Representative Simulated Trial Path Selection

Selection element	Governed value
Scenario Id	OS_NULL_REFERENCE
Scenario Seed Binding	2026081401
Selection Seed	2026081490
Search Replicates	5000
Batch Size	500
Roles	reject: eligibility: one_sided_p_value < alpha; selection rule: first eligible trial in increasing search index; non reject: eligibility: one_sided_p_value >= 0.5; selection rule: first eligible trial in increasing search index; near alpha boundary: eligibility: estimable trial; selection rule: minimum absolute(one_sided_p_value - alpha), tie to smallest search index

Actual aggregate simulated trial paths are selected deterministically from the governed scenario using a separate selection seed and frozen search rules. They supplement aggregate operating characteristics and do not replace them.

Deterministic Edge-Case Verification Fixtures

Artificial separated-time and zero-event cases are retained only as software verification fixtures for log-rank direction, decisions, and non-estimable handling.

- Analytic one-sided null rejection probability benchmark.
- Scenario-derived aggregate reject, non-reject, and near-alpha trial examples selected by a frozen seed/search rule.
- Artificial positive, negative, and non-estimable log-rank edge fixtures, labelled as validation fixtures rather than representative trial paths.
- Event-time, censoring, accounting, and parameter-bound invariants.

Reproducibility, Verification, and Change Control

Observed Execution Environment

Component	Recorded identity
Python	3.12.13
NumPy	2.2.6
PyYAML	6.0.3

Component	Recorded identity
Floating-point dtype	float64
Random-number generator	numpy.random.PCG64
Dependency lock	requirements-ci.lock

The environment above is a post-run traceability record, not a prospective design element. The dependency lock, float64 arithmetic, and PCG64 generator are part of the reproducibility contract.

Artifact Bindings

Artifact	Identity
Governed protocol	config/simulation_protocol.yaml -- SHA-256 09798cd52adedd742a10f39266c67df0d9fe40b4c693912e4962b47601446b61
Authoritative scientific results	platform/simulation_operating_characteristics/simulation_oc_status.json -- SHA-256 16bfab6f9abc171fa52ddd9d6d7c5035a4502c37956851cc247c51b8bc3d4fb8
Scenario registry	da115050f9d3fb69202b7154814c0eb204852efbe99f9b6107e548419f3f7768
Simulation code	3f5de78d6b174def4a8b92e77d4cde7064f25477f4cb18be76a295cbe17cc3ac
Scientific output	46ca63556e65c649b17c7ace2b70447691fd50d621c84d630880bb988fdb643e

Reproduce with python3 platform/simulation_precision.py, then run python3 platform/build_simulation_report.py. Identical governed inputs and seeds must reproduce identical scientific JSON and reports.

Qualification and Claim Boundary

- Classification: NON_MIDD_NON_CONFIRMATORY_DATA_FREE_METHODS_EVALUATION
- Model Influence: LOW
- Evidence Status: INFORMATIONAL_ONLY
- Authoritative Patient Data Used: No
- External Validation Completed: No

Reconstructed or synthetic CbzP data remain illustrative. This plan does not establish a clinically justified minimum effect, validate virtual patients against source IPD, support a clinical or filing decision, or convert TROPIC into a regulatory submission.

References and Governing Basis

- TROPIC simulation-precision research basis
- ICH M15: General Principles for Model-Informed Drug Development
- ICH E9(R1): Estimands and Sensitivity Analysis
- FDA: Adaptive Designs for Clinical Trials of Drugs and Biologics
- FDA: Complex Innovative Trial Design Meeting Program
- ADEMP framework
- OCTAVE framework